

GYA action plan: aligning research assessment practices with CoARA

1. Introduction

The Global Young Academy (GYA), established in 2010, brings together approximately [200 members](#) from six continents to amplify the voice of young scientists in evidence-informed and inclusive decision-making at global, regional and national levels. Members are selected by an internal selection committee based on their scientific excellence and commitment to societal engagement, serving five-year terms.

In 2023, the GYA signed the Coalition for Advancing Research Assessment (CoARA) agreement, committing to actively work towards reforming the global research assessment system and welcoming the initiatives arising from CoARA. This Action Plan outlines the strategies and activities that the GYA will implement to reform and enhance the member evaluation system, in line with the principles of quality, impact, diversity, inclusion, and integrity that underpin GYA's mission and are also integral to the CoARA agreement.

Guiding Principles. The principles guiding this Action Plan for improving research evaluation practices within the GYA and aligning with the CoARA agreement are as follows:

1. **Context Matters.** As we are a global organization characterized by internationality and interdisciplinarity, understanding the context of each candidate is essential for a fair evaluation. In this sense, both qualitative evaluation and reviewer training are key components of a successful evaluation. In this action plan we will focus on improving both dimensions of the GYA's selection processes.
2. **Quality is Multidimensional.** As an organization that aspires to provide solutions to society's intricate challenges, we value the importance of diverse contributions to research. We aim at emphasizing the recognition of the diversity of research contributions and careers in research, according to the needs and nature of the research. Also, we will explicitly mention in the call for applications and in the Reviewer Package the abandonment of

inappropriate uses in research evaluation of journal and publication-based metrics.

Objectives of the Action Plan

1. Generate evidence that can improve our scientific culture.
2. Foster dialogue about research evaluation practices.
3. Implement changes to membership assessment criteria aligned with the CoARA agreement.
4. Encourage a culture that values and rewards peer review and the diverse roles in research.

2. Overview of relevant GYA operations

As an academy, the GYA has a variety of activities that are not directly concerned with research assessment.

GYA's main activity of research assessment is the annual selection of new members, which is the focus of this document. Further, some more activities may include an assessment of a researcher or a research project, though in a small-scale setting:

- The selection of a group of GYA members for the Sasha Kagansky Interdisciplinary Grant, a small research grant (€ 10,000)
- Nominations of individuals for prizes, honors or special conferences (e.g., Lindau Nobel Meetings)
- Nominations of individual members to represent the GYA in related bodies, external committees, etc. (e.g. Expert committee for the ISC)

2.1. Responsible Parties for selection procedures

Work in the GYA is organized in committees and working groups. Committees oversee different aspects of the internal operations of the GYA, for example, its funding strategies, the legal framework for its operations, the elections of the GYA executive committee, and the selection of new GYA members. A working group focuses on a specific topic, for which it develops activities and output.

With regard to selection procedures, our small-scale internal selections and nominations are done by a *Nominations and Opportunities Committee*,

sometimes in cooperation with the Executive Committee. The selection of new members - our main topic - is dealt with by the Membership Selection Committee, supported by disciplinary panels and the Scientific Excellence Working Group. The next section gives details.

2.2. Overview of the current membership selection process

Membership selection takes place once per year and is based on applications. The GYA does not accept direct nominations from other Academies, though of course senior academy members may encourage a candidate to apply and provide a recommendation letter for them.

Applications are first screened for formal membership eligibility criteria and then assigned three independent reviewers. Completed evaluations are then assessed by a disciplinary panel. The disciplinary panel draws up a ranking and top candidates are forwarded to the final stage, where all disciplinary panels meet and take a decision together.

We currently run four disciplinary panels:

1. Panel of Arts & Humanities and Social Sciences: Arts, History, Linguistics & literature, Philosophy, Theology, Communication & media studies, Anthropology, Archaeology, Economics, Human geography, Law, Political science, Psychology, Sociology (including gender studies), and Education & pedagogy.
2. Panel of Natural Sciences: Biology, Chemistry, Earth and Environmental sciences, Space sciences and Physics.
3. Panel of Formal Sciences and Engineering: Computer science, Mathematics, Statistics, Engineering and Technology (including Information Technology).
4. Panel of Applied Sciences and Health: Medicine & health and Built Environment (including Land economy and economics, Urban and regional planning, Building construction and management, Architecture).

3. Overview of actions

In this section, we sketch our operational action plan for a 5-year time frame.

Our action plan has three parts:

Part 1: Exchange practices and experiences to enable mutual learning within and beyond the Coalition.

This part contains the following actions:

- a) Participation in CoARA WGs
- b) Organizing seminars and workshops to share findings and foster dialogue about research evaluation practices
- c) Expand CoARA in Latin America and Africa

Part 2: Improving our selection processes

This part contains the following actions:

- a) Adapt the call for new members
- b) Revise reviewer selection
- c) Develop a Reviewer Training Programme
- d) Develop a process for small selection procedures

Part 3: Evaluate criteria based on solid evidence in research on research, and make data openly available for evidence gathering and research

This part contains the following actions:

- a) Analyzing Global Views of Institutional Promotion Criteria
- b) GYA-IAP-ISC Initiative on Research Evaluation

3.1. Current status on our actions

3.1.1. Part 1: Exchange practices and experiences to enable mutual learning within and beyond the Coalition.

a) Participation in CoARA WGs:

The GYA maintains an active participation in several CoARA working groups, and wishes to expand our activities within CoARA in the forthcoming years.

Currently, we have representatives in the following working groups:

- Assessment of research proposals
- Reforming Academic Career Assessment

- Early-and-mid-Career Researchers (EMCRs) – Assessment and Research Culture
- Multilingualism and language bias in research assessment

b) Organizing seminars and workshops to share findings and foster dialogue about research evaluation practices

The GYA organized a panel on research assessment during its annual meeting and conference in Washington, D.C. in May 2024.

Furthermore, we are organizing a panel discussion at the World Science Forum 2024 in Budapest.

We are in the process of organizing a workshop at the ISC General Assembly meeting (Oman) and a panel discussion during the GYA Annual Conference of Young Scientists in Hyderabad (India).

c) Expand CoARA in Latin America and Africa

The GYA has aligned with local partner institutions to try to expand CoARA activities in Latin America.

This activity serves the following aims:

- Strengthen institutional capacities for purpose-oriented evaluation that recognises diversity of empowered responsible actors through the exchange of experiences and best practices.
- Systematize new and purpose-driven research evaluation practices that are adaptable to the realities of Latin American institutions and their ecosystems, centered on values that are of high relevance for their needs, including: transparency, ethical principles, quality, inclusivity, equity, and respect for institutional autonomy.
- Promote collaboration among researchers, managers, and evaluators to strengthen the research evaluation system and enhance the development and quality of the region's scientific output.

We are currently in the process of seeking funding for this activity.

In addition, we are currently researching the assessment of membership selection in young academies in Africa and plan to make a series of round table discussions across African Young and Senior academies to introduce the reform of research assessment, learn from the context and needs and together propose actions for reform.

3.1.2. Part 2: Improving our selection processes

At the core of our action plan is the review and update of the criteria for new membership selection to ensure they align with the ARRA's guiding principle.

In order to ensure a fair and more inclusive review of applications, we overhauled our operating procedures for the peer review of applications. This included

- i) making the peer review more inclusive,
- ii) the introduction of automated, unbiased pairing of applications and reviewers,
- iii) the development of a comprehensive review training programme,
- iv) setting up panels for the various disciplines to overview and streamline the review process, and
- v) taking into consideration global, disciplinary and gender parity tendencies in research participation in order to foster a global, diverse and equal community in the GYA.

To make the membership selection process more inclusive, we have changed our reviewer recruitment practices, to ensure diversity among reviewers and not to overburden reviewers with too many applications.

We have already implemented an algorithm that ensures an automated, unbiased pairing of applications and reviewers. This also had a beneficial effect on the workload of the membership selection committee. Regarding the other action items, we will move in two lines of actions:

a) Adapt the call for new members to be in line with the above guiding principle.

The appendix gives some examples on how we have applied the CoARA principles in our membership criteria.

We act along the following lines:

- 1) Making sure that we evaluate candidates following the principles, mainly including diversity of career paths and outputs, ensuring equal opportunity (context dependent evaluations) and avoiding the use of quantitative indicators.
- 2) Make the process more inclusive and ensure that we train all reviewers in a peer review process that adheres to GYA values and CoARA principles.

For this, we have implemented, for the first time for 2023, a partial automation of some parts of the selection process. Furthermore, reviewer guidelines have been developed, and the selection panels have been restructured to represent not only disciplinary, but also geographic variety.

- 3) Our future goal is to foster a global, diverse and equal community, by accounting for global, disciplinary and gender parity tendencies in research participation, with the help of evidence-based diversity criteria.

b) Develop a Reviewer Training Programme (Reviewer Package)

This action is important to ensure that all reviewers understand what these guiding principles are and how to recognise them when assessing GYA member candidates.

In order to ensure that the peer review of applications adheres to the CoARA principles and all reviewers are aware of these principles, the GYA has prepared a comprehensive reviewer training package. This includes a detailed evaluation guide based on CoARA principles, a separate short guide that sums up the main points (executive summary) which is meant to be used as a quick reminder after studying the detailed guide, a short video summary of the main points for reviewers who are auditory learners, and a detailed scoring guide that applies the CoARA principles.

Part 3: Evaluate criteria based on solid evidence in research on research, and make data openly available for evidence gathering and research

Action item a) Analyzing Global Views of Institutional Promotion Criteria

This action item consists in the preparation of a global study of research evaluation practices, compiling comparable datasets on a global scale. This study focuses on the criteria required to attain the status of full professor, with the ultimate goal of proposing evidence-based recommendations to enhance current evaluation practices and improve the overall research culture.

Action item b) GYA-IAP-ISC Initiative on Research Evaluation

The GYA has partnered with the InterAcademy Partnership (IAP) and the International Science Council (ISC) to study and discuss how to cooperate and work together towards a reform of research assessment at the international level. The work has mostly focused on the membership and network of the three organizations, though further outreach developments would be welcomed by the GYA. So far, the three organizations have published a synthesis report of the work undertaken by an ad hoc Scoping Group (available here: <https://www.interacademies.org/publication/future-research-evaluation-synthesis-current-debates-and-developments>) as well as a report on the state of the art of research evaluation at national academies (available here <https://globalyoungacademy.net/publications/snapshots-of-reform-researcher-evaluation-within-science-organizations/>).

4. Action timeline

2023

- Part 1 a) GYA sends representatives to several CoARA working groups
- Part 1 b) A first seminar to share findings and foster dialogue about research evaluation practices is prepared and scheduled to take place at the 2024 Conference of Young Scientists / GYA annual meeting.
- Part 2 a) A partial automation of some parts of the selection process is implemented. Furthermore, reviewer guidelines have been developed, and the selection panels restructured
- Part 3 a) Data collection for the project, conceptualization of the recoding of relevant variables, first draft of a report
- Part 3 b) Creation of a joint Scoping Group, interview with stakeholders from the various world regions, desk research, and publication of a first report

2024

- Part 1 c) First efforts to expand CoARA activities in Latin America.
- Part 2 a) The application form is reworked to align with CoARA principles and implemented for the 2024 membership evaluation
- Part 2 b) The reviewer training package is developed and tested for the first time for the 2024 membership evaluation
- Part 3 a) Submission to a scientific journal, several rounds of peer review and revisions.
- Part 3 b) Several rounds of discussion with an external consultancy that undertook both a survey of national academies and 12 in-depth interviews; publication of a second report

2025

Part 1 b) workshop at the ISC General Assembly meeting (Oman) and a panel discussion during the GYA annual conference of young scientists in Hyderabad (India).

Part 2 a) First evaluation of the changes introduced in the membership selection process. Candidate survey. Comparison of cohort characteristics with respect to previous cohorts (diversity).

Part 2 b) First evaluation of the changes introduced in the Reviewer Training Programme. Survey of reviewers.

Parts 2 a and b) Evaluation of the necessary adjustments on the basis of this follow-up for the selection of members call 2025.

Part 2 c) Develop a process for small selection procedures. Decide on a first call in which to test the applied changes and monitor the outcome of the implemented changes.

2026

Part 2 a and b) Follow up of the implemented changes. Permanent evaluation of possible adjustments.

Part 2 c) Expand to other small selection procedures. Monitor the outcome of the implemented changes.

2027

Part 2 a, b and c) Follow up of the implemented changes. Permanent evaluation of possible adjustments.

5. Appendix

In the assessment of GYA membership selection criteria such as scientific excellence and commitment to society, we have applied the CoARA principles through (i) clear communication of our commitment to CoARA principles, (ii) the way we phrase questions and (iii) the way we evaluate application documents.

Example on Point 1

An introductory text preceding the questions relating to scientific excellence explains that GYA is a strong supporter of the CoARA principles, and therefore it is committed to avoiding the inappropriate use of quantitative metrics. Applicants are directed to a FAQ question where we explain in more detail what the acronym CoARA stands for and what principles the coalition has adopted in the evaluation of research performance.

“FAQ 18. What are the CoARA principles that the GYA uses in the evaluation of applications?

The GYA uses the CoARA (Coalition for Advancing Research Assessment) principles to evaluate applications. These principles emphasize the importance of responsible research assessment, including transparency, diversity of outputs, and the value of contributions to both science and society.”

The introductory text makes it explicit that just referring to publications or citations in the essays is insufficient, and achievements are not measured in terms of securing research grants and funding either. We instruct applicants to refrain from referring to their bibliometric indicators or h-index. The text makes it explicit that in the essays we are looking for a presentation of the quality and rigor of the applicant's work, and a demonstration of their ability

to further advance knowledge and innovation. We invite candidates to explain the relevance of their research findings for their field and beyond, and to showcase the difference their work made at the national, regional, or international level. We instruct applicants to describe how they acquired the skills they have, and how they contributed to supporting their research community.

Example on point 2

Q A3.

“What are your distinctive skills, and how did you acquire them? (2200 characters)”

This question further steers applicants away from metrics, and places emphasis on professional growth in terms of skills.

Q A4.

“What are your major achievements as a researcher? What did you find, what difference did it make, and why is this relevant? Please explicitly include three (3) most significant research outputs and outcomes (of any kind). (2200 characters)”

This question about achievements invites applicants to reflect on the difference they made and the relevance of their work inside and outside of their community. In observance of the CoARA agreement, it avoids reference to any metrical indicators. A related FAQ question explains that research outputs and outcomes come in many forms, from publications to real-world applications and impact on societal well-being. We emphasize that the GYA is aware that what counts as a significant output or outcome is context and field dependent, and we highlight the importance of FAIR principles.

Our take on CVs

Our application documents require a 4-page CV. The online application form emphasizes again that the GYA is not looking for a de-contextualized list of qualitative metrics:

“Please ensure that these documents *do not contain any reference to bibliometrics (e.g. citation counts or h-index), university ranking data, or income from grants or patents*. Do not provide a list of publications. We will ignore all documents containing such information.”

In addition, CVs are encouraged to be at least partially narrative, in order to put achievements in context:

“Your CV should contain a brief education and employment history and make your achievements clear to a general audience. For example, a few words summarizing your most significant publications are more useful than a complete list of every paper that you have authored.”